# SAMPLE ROUTING MODULE (SRM)

## MODULAR EMBEDDED COMMUNICATION SYSTEM

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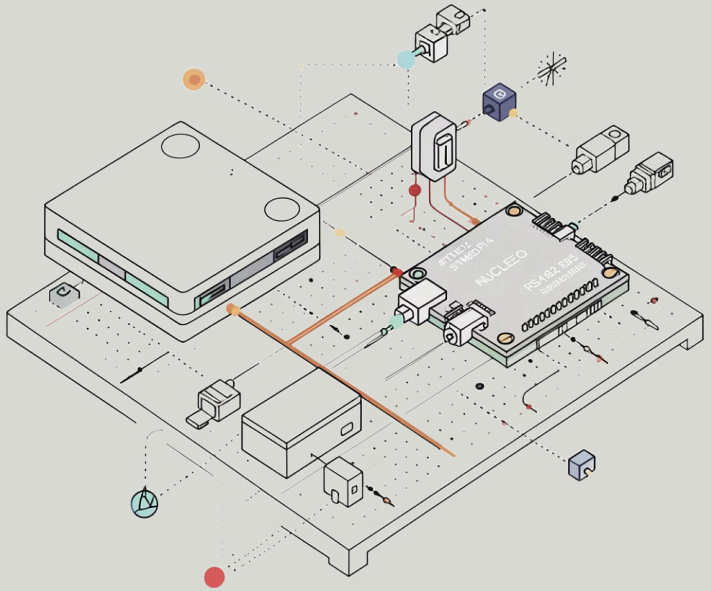
Embedded C/C++ Course Project

Gamma Intelligence OÜ

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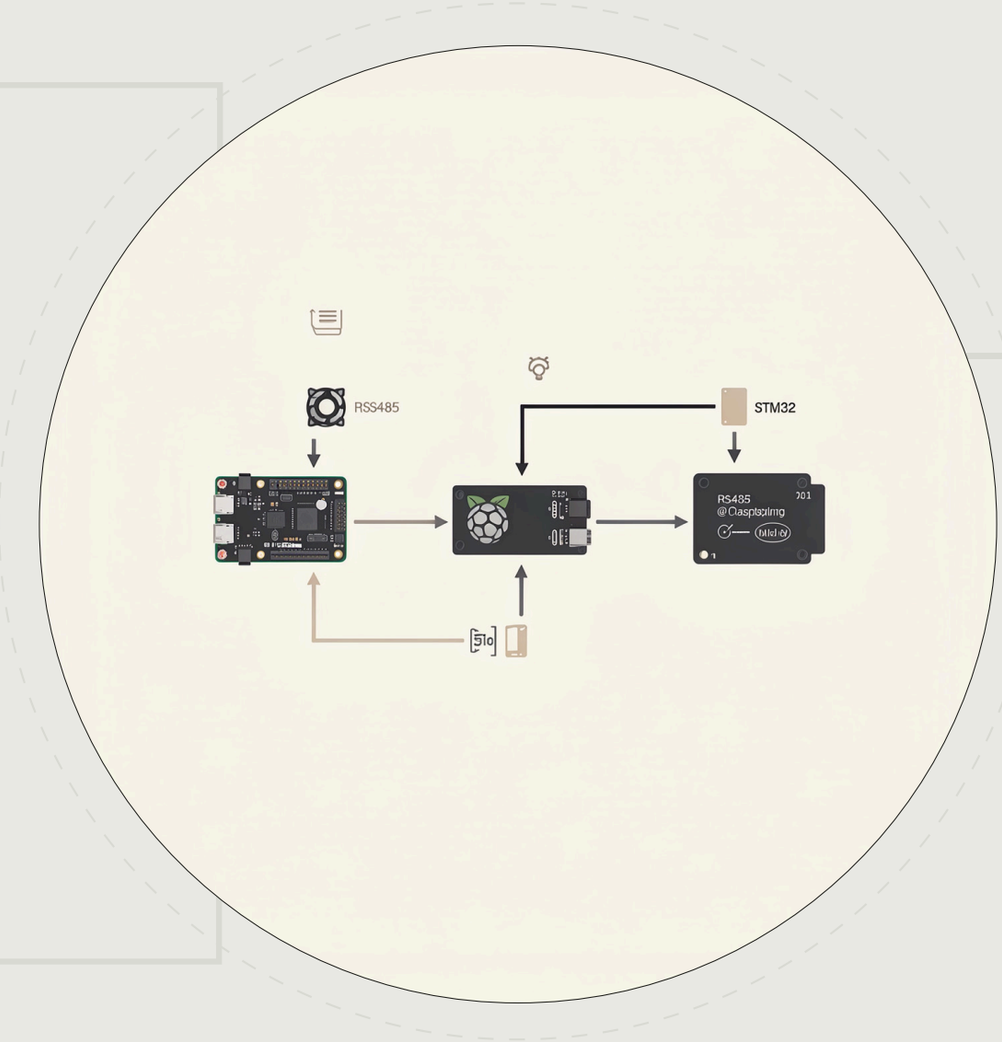
# PROJECT GOAL

The Sample Routing Module (SRM) simulates a small industrial or medical machine using a master-slave architecture. The Raspberry Pi serves as the master controller, while the STM32 microcontroller acts as the slave device. Communication occurs via RS485 protocol, demonstrating a scalable embedded architecture suitable for industrial automation applications.

## MASTER

Raspberry Pi invia comandi




## BUS RS485

Collegamento seriale differenziale

## SLAVE

STM32 esegue istruzioni

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# HARDWARE COMPONENTS

### RASPBERRY PI 4 MODEL B

Master controller with Linux OS

### STM32 NUCLEO-F207ZG

ARM Cortex-M3 slave processor

### RS485 TRANSCEIVER MODULE

Differential serial communication

### LCD1602 I2C DISPLAY

16×2 character operator feedback

### ULN2003 STEPPER DRIVER

Motor control IC module

### 28BYJ-48 STEPPER MOTOR

5V unipolar stepper motor

Additional components include USB-to-RS485 converter, jumper wires, and breadboard for prototyping.

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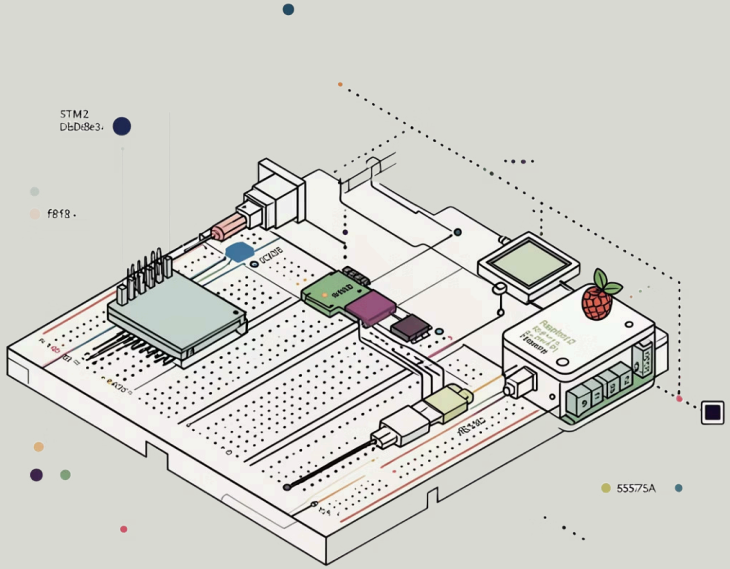
# WIRING CONNECTIONS

## STM32 PINOUT

- **PB6** → RS485 TX
- **PB7** → RS485 RX
- **PB8** → LCD SCL
- **PB9** → LCD SDA
- **PD4** → ULN2003 IN1
- **PD5** → ULN2003 IN2
- **PD6** → ULN2003 IN3
- **PD7** → ULN2003 IN4
- **PA5** → Onboard LED

## POWER & GROUND

LCD1602 and ULN2003 connected to 5V power supply and common ground. RS485 module requires separate 5V connection for transceiver operation.



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# WHY RS485?

### SUPERIOR NOISE IMMUNITY

Differential signaling rejects common-mode noise in industrial environments

### LONG CABLE RUNS

Supports distances up to 1200 meters between devices

### MULTI-DEVICE BUS

Daisy-chain multiple slaves on single communication line

### INDUSTRIAL STANDARD

Common in medical equipment and factory automation systems

### UART DIRECT CONNECTION

Short distance only, single device, susceptible to noise

### RS485 DIFFERENTIAL BUS

Long distance, multi-device, noise-resistant differential pairs

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# SRM PROTOCOL FRAME

Custom binary protocol designed for reliable industrial communication with error detection and addressing capabilities.

1 **SOF**
Start of Frame

2 **VER**
Protocol Version

3 **FLAGS**
Control Flags

4 **SRC/DST**
Addressing

5 **CMD**
Command Type

6 **SEQ**
Sequence Number

7 **LEN**
Payload Size

8 **PAYLOAD**
Data Field

9 **CRC16**
Validation Check

## EXAMPLE FRAME

AA 01 00 10 01 30 05 04 54 45 53 54 3F 92 - PING command from address 0x10 to 0x01 with "TEST" payload

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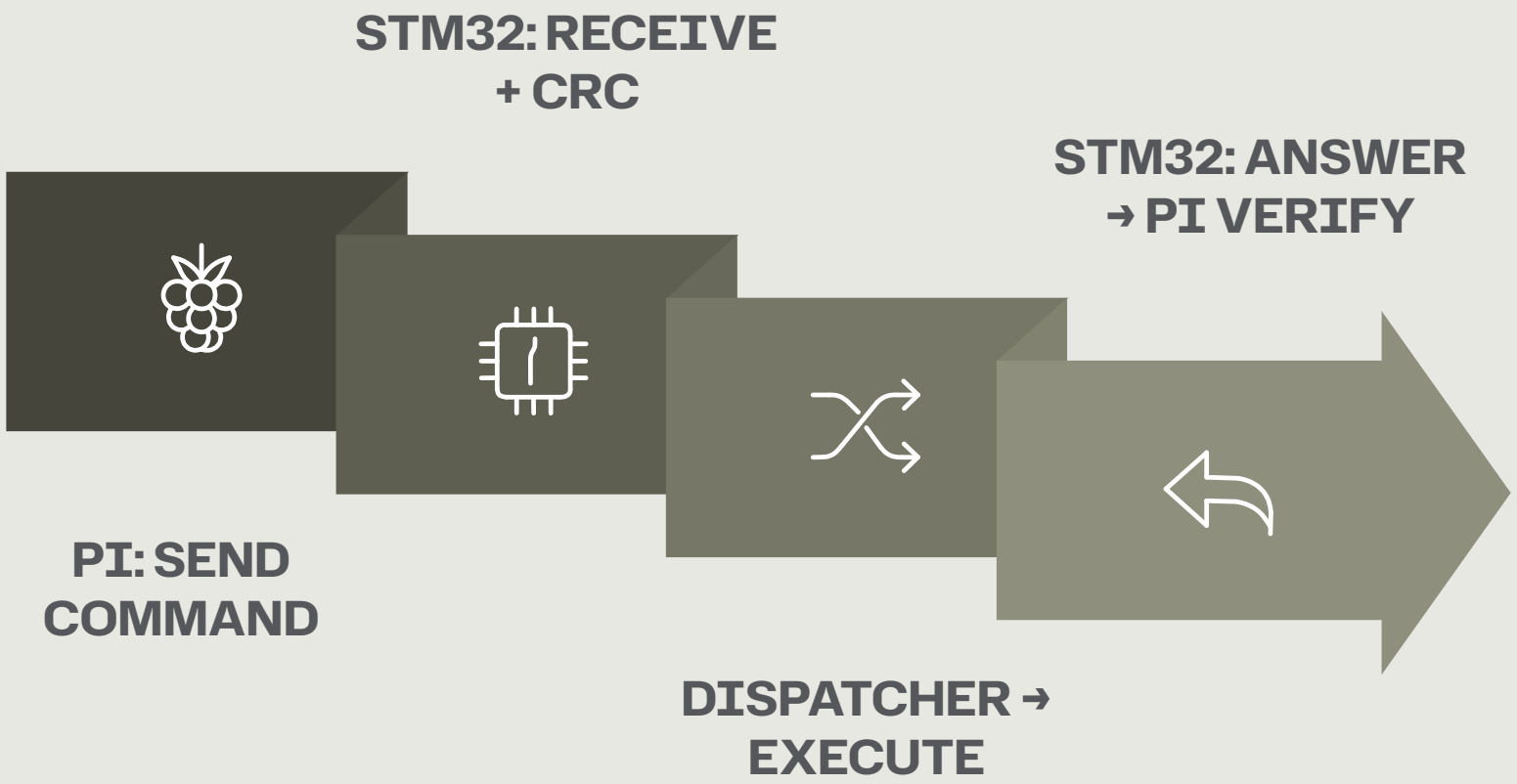
# COMMAND FLOW




01

**RASPBERRY PI SENDS COMMAND FRAME**

02

**STM32 RECEIVES COMPLETE FRAME**

03

**CRC16 VALIDATION CHECK**

04

**COMMAND DISPATCHER ROUTES REQUEST**

05

**PERIPHERAL EXECUTES ACTION**

06

**RESPONSE FRAME GENERATED**

07

**RASPBERRY VALIDATES RESPONSE**

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# SUPPORTED COMMANDS

| Command | Purpose | Action |
|---|---|---|
| PING | Test connection | Return PONG response |
| LED ON | Activate onboard LED | PA5 high, LCD shows "LED ON" |
| LED OFF | Deactivate LED | PA5 low, LCD shows "LED OFF" |
| STATUS | System status | Return system state and LCD message |
| LCD CLEAR | Clear display | Reset LCD, show "READY" |
| LCD WRITE L1 | Line 1 text | Update first LCD line |
| LCD WRITE L2 | Line 2 text | Update second LCD line |
| STEPPER CW | Rotate clockwise | Motor CW, LCD shows "MOTOR CW" |
| STEPPER CCW | Rotate counterclockwise | Motor CCW, LCD shows "MOTOR CCW" |
| STEPPER HOME | Return to origin | Software return, LCD shows "MOTOR HOME" |

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# LCD FEEDBACK INTERFACE

### READY

System initialization complete, awaiting commands

### PONG

PING command response received

### LED ON / OFF

Onboard LED state change confirmation

### MOTOR CW / CCW

Stepper motor rotation direction active

### MOTOR HOME

Software home position return initiated

The LCD1602 provides operator feedback in machine-style format, displaying current state and command execution results.

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# STEPPER MOTOR INTEGRATION

## HARDWARE SETUP

The ULN2003 stepper driver interfaces the STM32 GPIO pins to the 28BYJ-48 5V stepper motor. Four output pins control motor coils in sequence for precise rotation.

## SOFTWARE CONTROL

CW and CCW commands rotate motor in respective directions. HOME command implements software return to origin position. Future improvement: hardware end-stop switch for accurate homing.

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